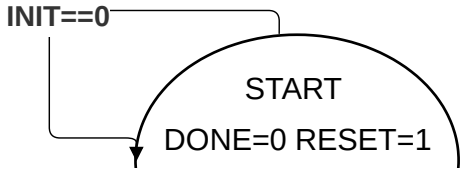
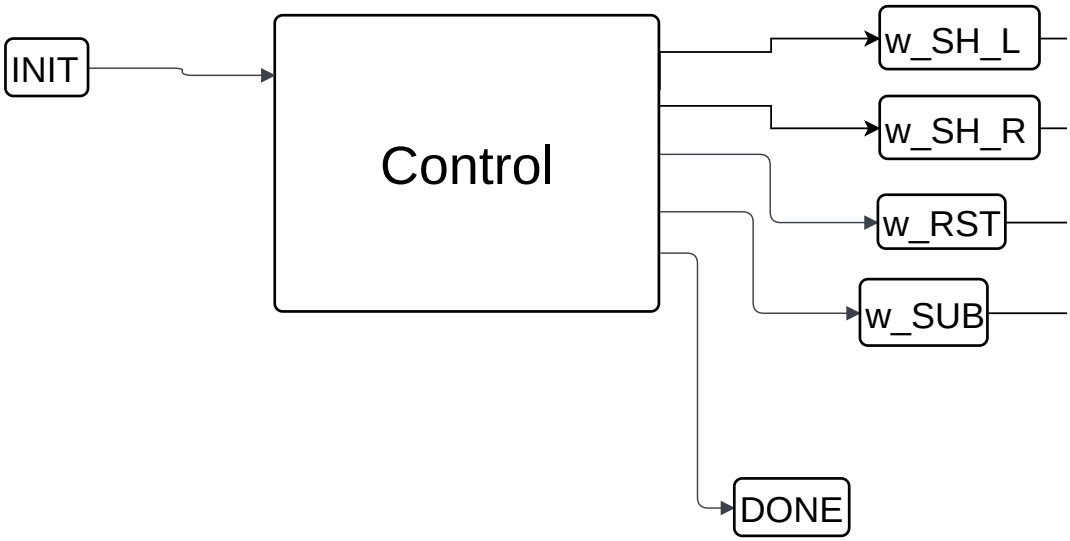
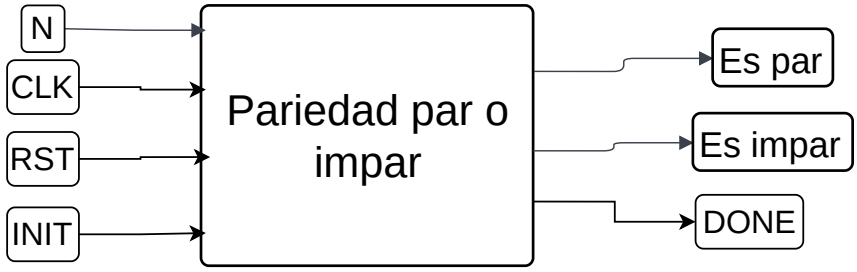
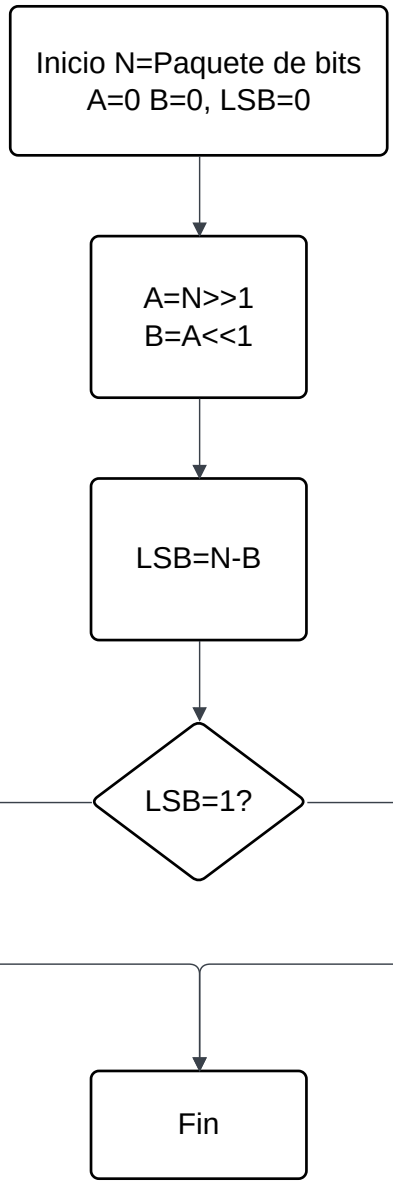
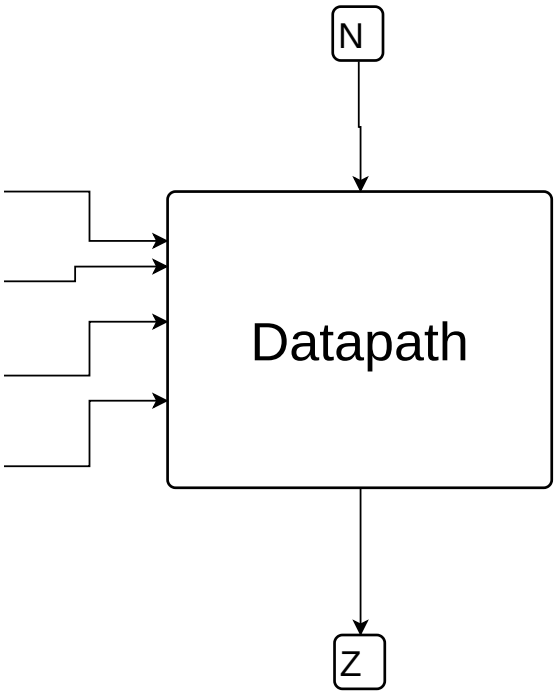
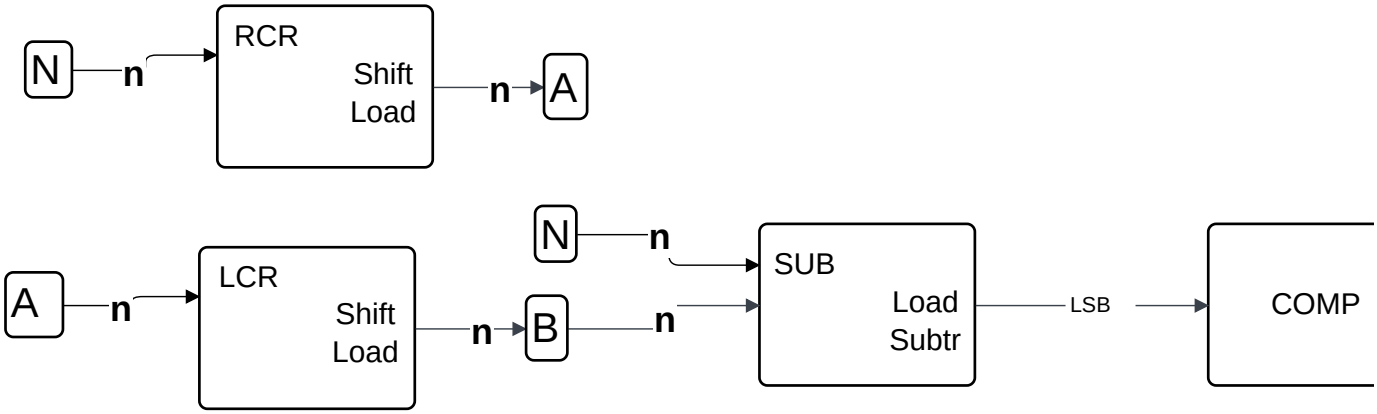

INIT==0
START
DONE=0 RESET=1

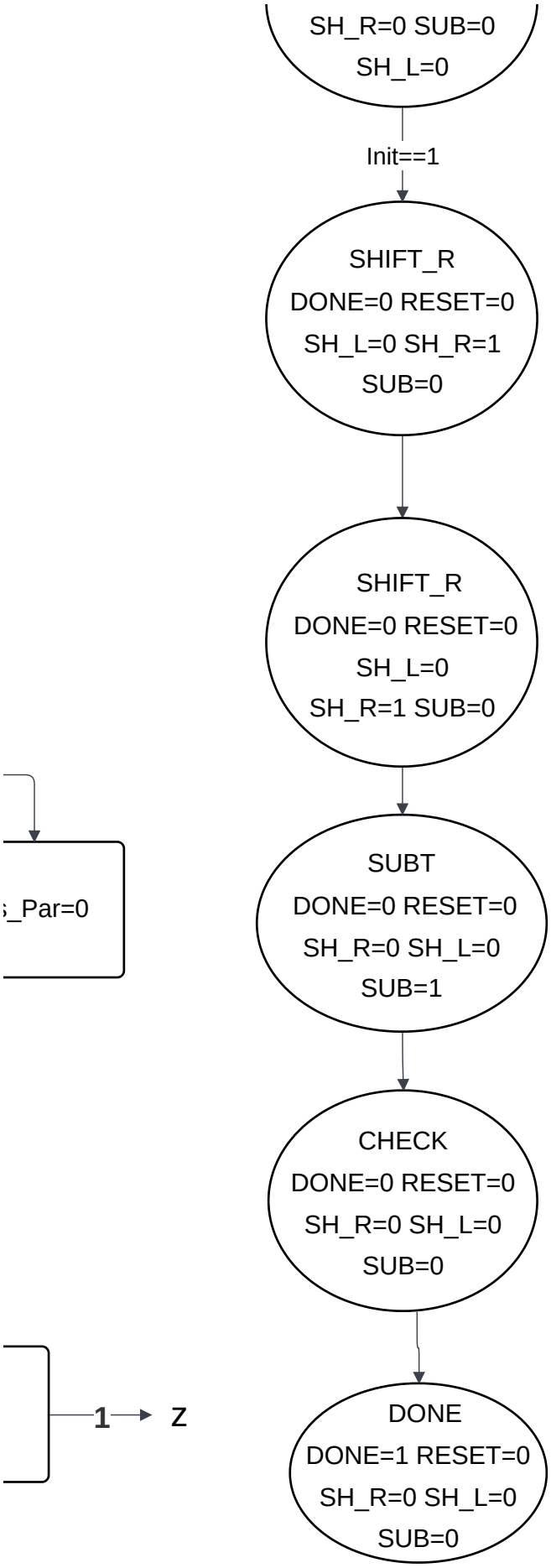

INIT
Control
w_SH_L
w_SH_R
w_RST
w_SUB
DONE


N
CLK
RST
INIT
Pariedad par o impar
Es par
Es impar
DONE


Inicio N=Paquete de bits
A=0 B=0, LSB=0
A=N>>1
B=A<<1
LSB=N-B
LSB=1?
Sí
No
Es_Impar=1
Es
Fin


N
Datapath
Z


N
n
RCR
Shift
Load
n
A
A
n
LCR
Shift
Load
n
B
N
n
n
SUB
Load
Subtr
LSB
COMP


SH_R=0 SUB=0
SH_L=0
Init==1
SHIFT_R
DONE=0 RESET=0
SH_L=0 SH_R=1
SUB=0
SHIFT_R
DONE=0 RESET=0
SH_L=0
SH_R=1 SUB=0
_Par=0
SUBT
DONE=0 RESET=0
SH_R=0 SH_L=0
SUB=1
CHECK
DONE=0 RESET=0
SH_R=0 SH_L=0
SUB=0
1
Z
DONE
DONE=1 RESET=0
SH_R=0 SH_L=0
SUB=0